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Review

# Is sentinel lymph node biopsy warranted for desmoplastic melanoma? A systematic review

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**Summary** *Background:* Desmoplastic melanoma (DM) is an uncommon malignancy associated with a high local recurrence rate. The aim of this systematic review was to determine the positivity rate of sentinel lymph node biopsy (SLNB) in patients with DM. The secondary outcome was to establish if SLNB is warranted for both pure DM (PDM) and mixed DM (MDM).
*Methods:* A full systematic literature review of SLNB in DM was performed by two authors in January 2016. Ovid MEDLINE, Ovid EMBASE and the Cochrane Central Register of Controlled Trials were searched.
*Results:* Sixteen studies involving 1519 patients having SLNB in DM were included, of which 99 patients had positive SLNB (6.5%). Two articles reported a significantly reduced disease-free survival (DFS) with positive SLNB and three published a reduced melanoma-specific survival (MSS). Six studies compared SLNB in MDM and PDM. Of the 275 patients, 38 (13.8%) had a positive SLNB in MDM compared to 17 of 313 patients (5.4%) with positive SLNB in PDM.
*Conclusions:* Rates of positive SLNB in DM are reduced compared to other variants of melanoma; however, nodal status may still predict DFS and MSS. MDM is associated with a higher rate of micro-metastases to regional lymph nodes than PDM, and DFS and MSS may be lesser in MDM than in PDM. We would recommend the consideration of SLNB in MDM. However, with

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such low rates of positive SLNB in PDM, and in the absence of high-risk features to stratify patients, we would not recommend SLNB in PDM.
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## Introduction

Desmoplastic melanoma (DM) is an uncommon malignancy representing 1%—3% of all cutaneous malignant melanomas, lacking the typical features of commoner variants of melanoma.[1,2] It is histologically characterised by spindle-shaped melanocytes within a fibro-collagenous (desmoplastic) stroma and has a predilection for the head and neck in older male patients.[3] It presents as a thicker tumour than non-DMs and is associated with a high local recurrence rate.[2—5]

Despite patients presenting with thicker tumours, regional metastases are suggested to be lower than in other melanomas, and DM may be associated with an improved overall survival.[2—4,6] Estimates of rates of positive sentinel lymph node biopsy (SLNB) have been reported as 0%—18%.[7—10]

In 2004, Busam et al. distinguished pure DM (PDM), with desmoplasia present throughout $\geq$90% of the tumour, from combined DM, with $<$90% desmoplasia present.[11] The combined variant was renamed 'mixed' in 2005,[12] and early data suggested that PDM metastasised less frequently to regional nodes than mixed DM (MDM), with a longer disease-free survival (DFS).[10,12—14] However, recent studies argue that the incidence of micro-metastases to regional lymph nodes in PDM is higher than previously reported, and SLNB is warranted for all cases of DM.[7,9,15] It has led to the publishing of conflicting national SLNB guidance in the United States.[16,17]

The aim of this systematic review was to determine the positivity rate of SLNB in patients with DM. The secondary outcome was to establish if SLNB is warranted for both PDM and MDM.

## Methods and materials

### Search strategy

Two authors (JAD and JCRW) independently conducted a full systematic literature review of all studies in Ovid MEDLINE, Ovid EMBASE and the Cochrane Central Register of Controlled Trials. Both 'free-text term' and 'MeSH term' searches were performed, using variations of the keyword 'desmoplas*'. Keywords were combined using Boolean operators. Articles in the English language were considered. Each author's search results were merged and duplicate citations were discarded. Titles and abstracts were screened, and studies unrelated to the research objective were discarded. The full text of the relevant papers was retrieved and examined by each author independently for consideration for inclusion/exclusion. The final list of the included studies was compared and discussed between the two authors (JAD and JCRW). The reference lists of all identified articles were reviewed for relevant studies. Finally, the archives of pertinent journals were hand-searched for relevant articles (Journal of Plastic, Reconstructive and Aesthetic Surgery; Plastic and Reconstructive Surgery; Annals of Surgical Oncology; and European Journal of Surgical Oncology). The authors of all included studies were contacted to identify potential further or ongoing research. The search was last undertaken in January 2016. Data were independently extracted by the two authors (JA and JCRW) using a standardised form. The data were then cross-checked between the two authors to confirm the accuracy of extraction.

### Selection criteria

The inclusion and exclusion criteria were decided during the protocol stage of the review. All papers reporting on outcomes of patients with DM as a primary outcome were included for review to optimise population homogeneity. Data were extracted from studies reporting the incidence of positive lymph nodes in SNLB for DM. Studies not reporting SLNB outcomes were excluded. Studies including SLNB with other lymph node examinations (microscopic and macroscopic disease) in which the positivity rate for SLNB

could not be determined were excluded.[14] Case reports, review articles and duplicate patient populations were excluded. Where two studies used the same database or patient population and the time period was over-lapping, there is a risk of duplication bias; in such cases, the most extensive data collection was used, with the shorter study excluded. Where two studies had similar time periods from the same database, however, one restricted participants by anatomical site, both were included.

The study selection process was in accordance with the PRISMA statement and is reported in Figure 1. The search strategy identified 232 papers, which were screened for retrieval, and 191 were deemed irrelevant based on their title or abstract. Forty-one studies were retrieved for detailed review, of which 19 did not meet the inclusion criteria. Six further studies were excluded as they did not report SLNB outcomes. The remaining 16 papers were included in the review. The principal summary measure was the rate of positive SLNB in patients with DM, expressed as a percentage, and the secondary measure was the incidence of positive SLNB comparing PDM and MDM.

## Results

Sixteen studies were included in the review, and all were observational in design and conducted in three countries (Table 1).[1,3,7–10,13,15,18–25] Of the 16 studies, 10 included less than 50 patients and the average Breslow thickness was ≥2 mm in all articles. Two included only patients with head and neck DM, whereas the remainder included DM of all sites.[20,21] Almost all studies were retrospective case series (level IV evidence), except one case–control study (level 3 evidence).[1] There was considerable variability between the study periods of the included articles, ranging from 5 to 25 years. The follow-up periods were also heterogeneous (range 0.9–9.9 years). However, patient demographics were comparable between groups.

In total, 1519 patients were identified from all studies, and 99 SLNB (6.5%) were positive (Table 2). Five of the 16 studies did not identify a positive SLNB in any patient. Two articles reported a significantly reduced DFS with a positive SLNB, and three published a reduced melanoma-specific survival (MSS). Egger et al. reported a reduced overall

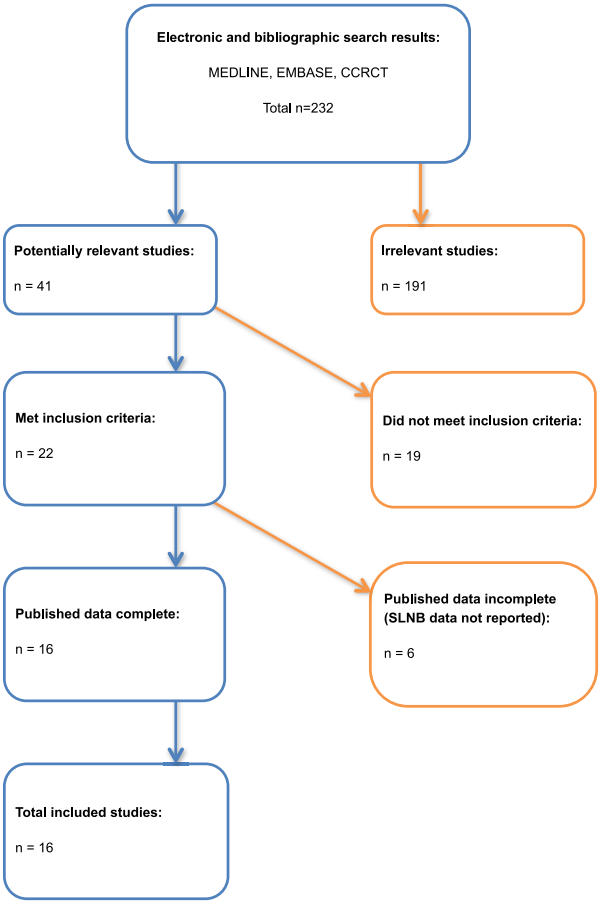


**Figure 1** Study attrition chart.

**Table 1** Demographic data of included studies.

| Study | Year | Study design | Country | Study period (years) | N | Median age (years) | M:F | DM site | Median follow-up (years) |
|---|---|---|---|---|---|---|---|---|---|
| Jaroszewski | 2001 | Case series | USA | 15 | 59 | 62.8 | 1.7:1 | All | 3.83[a] |
| Gyorki | 2003 | Case series | USA | 5 | 27 | 64 | 2.8:1 | All | 2.25 |
| Su | 2004 | Case series | USA | 5 | 33 | 61 | 3.1:1 | All | 1.66 |
| Livestro | 2005 | Case—control | USA | 32 | 89 | 63.9 | 1.7:1 | All | 4.3 |
| Pawlik | 2006 | Case series | USA | 11 | 65 | 61 | 1.3:1 | All | 2.9 |
| Posther | 2006 | Case series | USA | 23 | 129 | 55.2 | 1.7:1 | All | 5.1[a] |
| Thelmo | 2006 | Case series | USA | 6 | 16 | 57.5[a] | 2.2:1 | All | 0.9[a] |
| Cummins | 2007 | Case series | USA | 5 | 28 | 64 | 1:1 | All | 1.74 |
| Maurichi | 2009 | Case series | Italy | 25 | 242 | 64 | 1.3:1 | All | 9.9 |
| Murali | 2010 | Case series | Australia | 14 | 252 | 61 | 1.9:1 | All | 2.9 |
| Wasif | 2011 | Case series | USA | 18 | 1735 | 69 | 1.8:1 | All | 5[b] |
| Mohebati | 2012 | Case series | USA | 24 | 47 | 71 | 1.4:1 | Head and neck | 5 |
| Eppsteiner | 2012 | Case series | USA | 5 | 467 | 70[a] | 2.3:1 | Head and neck | 4.9[b] |
| Broer | 2013 | Case series | USA | 12 | 22 | 64 | 3:1 | All | 2.1[a] |
| Egger | 2013 | Case series | USA | 11 | 47 | 57 | 2.1:1 | All | 6.4 |
| Han | 2013 | Case series | USA | 18 | 205 | 66 | 2.2:1 | All | 6.3 |

[a] Mean.
[b] Longest recorded follow-up.

survival (OS) with an ulcerated primary DM and a positive SLNB.

Two studies compared positive SLNB in DM with non-DM. Livestro et al. reported a case—control study matched for age, sex, tumour thickness and year of diagnosis and demonstrated a positive SLNB rate of 8% in DM compared to 34% in non-DM. Pawlik et al. performed a similar comparison, albeit without matching, and reported a positive SLNB rate of 17.5% in non-DM compared to 6.2% in DM. However, when DM was divided into MDM and PDM, positive SLNB rates were 15.8% and 2.2%, respectively, with PDM significantly lower than MDM and non-DM ($P < 0.01$).

Six studies compared SLNB in MDM and PDM. Of a total of 275 patients in these studies, 38 (13.8%) had a positive SLNB in MDM compared to 17 of 313 patients (5.4%) with positive SLNB in PDM. Of these, Mohebati et al. did not identify a positive SLNB in either cohort. Two studies reported that MDM was associated with a reduced DFS. Pawlik et al. reported that 2.2% and 15.8% of the patients with PDM and MDM, respectively, had positive SLNB. All patients in this study had SLNB, and PDM was associated with a significantly improved 3-year DFS compared to the mixed subtype. All the 46 patients (100%) with PDM were free of disease at 3 years compared to 15 of 19 patients (79.1%) in the MDM cohort ($P = 0.004$).

Han et al. reported a high rate of melanoma-related deaths in those with a positive SLNB in PDM (2 of 6 patients; 33.3%) compared to those who had a negative SLNB (4 of 61 patients; 6.6%). For patients with MDM, 3 of 15 patients (20%) with a positive SLNB died of melanoma-related causes compared to 12 of 46 patients (26.1%) who had a negative SLNB.

Three studies reported a definition of DM that differed from the criteria published by Busam et al. and were more likely to select tumours with PDM.[11] The remaining studies reported DM as defined by Busam et al., or it was not defined. Two studies comparing PDM and MDM defined PDM as 80%—90% desmoplasia or greater compared to ≥90% in the remaining four studies.

## Discussion

Data from this systematic review provides further evidence that the prevalence of micro-metastases in DM is lower than in other variants of melanoma. A positive SLNB in DM may be associated with reduced DFS and MSS. In addition, positivity rates differ for the two histological subtypes, with PDM exhibiting lower rates of positive SLNB than MDM.

Early data suggested a poorer prognosis associated with DM compared to other types of melanoma; however, more recent studies have reported comparable or improved survival, particularly when matched for tumour thickness.[1,3,4,26] DM tumours are typically thicker on presentation; however, the relationship between survival and Breslow thickness is less clear than in other melanomas, with conflicting reports on the prediction of recurrence and survival.[1—3,12] Therefore, variation in the biology of DM has been proposed, evidenced by differences from other melanomas on microarray profiling, absence of activating BRAF mutations, high local recurrence rates of up to 40% and reduced rates of nodal metastasis.[1,3,5,27—30]

Suggested differences in the behaviour and rates of nodal metastasis in DM has led to varying opinions on the value of SLNB in this cohort compared to that in other variants of melanoma where it is an important prognostic indicator.[31] The overall rate of positive SLNB in the review was 6.5%, which was considerably lower than the positivity

**Table 2** SLNB outcomes for included studies.

| | Study | Median breslow thickness (mm) | Primary outcome | | Secondary outcome | | Positive SLNB association with DFS, MSS and OS |
|---|---|---|---|---|---|---|---|
| | | | SLNB patients (n) | SLNB positive (%) | MDM SLNB positive (n)/total cases (n) | PDM SLNB positive (n)/total cases (n) | |
| 1 | Jaroszewski 2001 | 6.5[a] | 12 | 0 (0%) | — | — | — |
| 2 | Gyorki 2003 | 2.2 | 24 | 0 (0%) | — | — | — |
| 3 | Su 2004 | 2.8 | 33 | 4 (12.1%) | — | — | — |
| 4 | Livestro 2005 | 2.6 | 25 | 2 (8%) | — | — | Reduced MSS |
| 5 | Pawlik 2006 | 2.9 | 65 | 4 (6.2%) | 3/19 (15.8%) | 1/46 (2.2%) | Reduced DFS for MDM vs PDM |
| 6 | Posther 2006 | 4.4[a] | 12 | 0 (0%) | — | — | — |
| 7 | Thelmo 2006 | 3.9[a] | 16 | 0 (0%) | — | — | — |
| 8 | Cummins 2007 | 2.3 | 15 | 1 (6.7%) | — | — | — |
| 9 | Maurichi 2009 | 1.9 MDM<br>2.1 PDM | 100 | 9 (9%) | 7/51 (13.7%) | 2/49 (4.1%) | — |
| 10 | Murali 2010 | 2 | 252 | 17 (6.7%) | 11/129 (8.5%) | 6/123 (4.9%) | Reduced DFS<br>Reduced DFS for MDM vs PDM |
| 11 | Wasif 2011 | 3.0[a] | 505 | 14 (2.8%) | — | — | — |
| 12 | Mohebati 2012 | 6.1[a] | 21 | 0 (0%) | 0/7 (0%) | 0/14 (0%) | — |
| 13 | Eppsteiner 2012 | 3.5 | 165 | 8 | — | — | Reduced MSS |
| 14 | Broer 2013 | 3.9[a] | 22 | 4 (18.2%) | 2/8 (25%) | 2/14 (14.3%) | — |
| 15 | Egger 2013 | 2.6 | 47 | 8 (17.0%) | — | — | Reduced DFS<br>With ulcerated primary, reduced OS |
| 16 | Han 2013 | 3.7 | 205 | 28 (13.7%) | 15/61 (24.6%) | 6/67 (9.0%) | Reduced MSS |
| | | **Total** | **1519** | **99 (6.5%)** | **38/275 (13.8%)** | **17/313 (5.4%)** | |

[a] Mean.

rate of 20% reported in Multicenter Selective Lymphadenectomy Trial-1 (MSLT-I) for all melanomas.[31] However, the positive SLNB rate for MDM was 13.8%, which is closer to the rate for all melanomas.

Criteria for which patients were offered SLNB were poorly reported in the studies in this review. Differences in positive SLNB rates published may in part be explained by small sample sizes and varying definitions of DM in studies prior to Busam et al.'s criteria, posing a risk of misclassification bias. Jaroszewski et al. defined DM as a significant invasive component composed of atypical spindle cells in a fibrotic stroma in a series of 12 patients having SLNB. Gyorki et al. similarly described DM as a prominent stromal fibrosis throughout the entire tumour predominantly composed of fusiform melanocytes constituting >90% of the tumour cell population in a cohort of 24 patients having SLNB.[3,8] It is likely that both the studies included a high proportion of PDM, which may account for the absence of positive SLNB observed. Although two studies comparing PDM and MDM defined PDM as 80%–90% desmoplasia throughout the tumour rather than >90% as defined in Busam et al.'s criteria, the likelihood of misclassification appears less as both report MDM as desmoplasia in <50% of the tumour, maintaining a distinction between the two.[7,13]

Although cumulative rates of positive SLNB in DM are reduced, nodal status may still predict DFS and MSS.[1,9,15,19,20] Offering SLNB in MDM with a positivity rate of 13.8% appears reasonable, although it is less clear with regards to the rate of 5.4% in PDM. MDM is characterised by a more aggressive course and associated with a worse DFS and MSS than in PDM.[10–12] Hawkins et al. reported a 5-year melanoma-specific mortality of 31% and 11% for MDM and PDM, respectively.[12] Han et al. reported that 2 of 6 patients (33%) with positive SLNB in PDM died of a melanoma-related cause compared to 3 of 15 patients (20%) with MDM[9]; this may suggest that PDM is aggressive in a small group of patients where micro-metastases are present.

SLNB should be offered on a routine basis after considering the benefits of prognostic information such as the surgical risks of a staging procedure and cost-effectiveness within a specified patient population. The National Institute for Health and Clinical Evidence in the United Kingdom has ceased to recommend SLNB in thin melanomas (≤1 mm Breslow thickness), in which positive SLNB rate is 8%–12% and complication rate is 10%.[31–33] DM typically presents as a thick tumour; however, studies in this review included some patients with tumours ≤1 mm Breslow thickness, although how many had SLNB is unclear. High-risk features have not been consistently reported in DM, and further research should be undertaken to establish if all patients with MDM should be offered SLNB based on stratification using clinicopathological factors. Analysing patients with PDM who are at the greatest risk of regional metastases and may warrant SLNB could improve outcomes in this small cohort.

Following a positive SLNB, national guidelines advise the consideration of completion lymph node dissection (CLND), with the aim of limiting further metastasis and providing regional control.[32] Two studies reported that up to a quarter of patients with DM may have non-sentinel lymph nodes identified on CLND histopathological examination, although the impact of CLND on DFS and MSS was not reported.[9,15,32]

Distant metastases may develop in 11%—40% of cases in DM, with a predilection for the lungs and brain.[2,3,6,29,30] Maurichi et al. reported the development of distant metastases in 24 patients (20%) with PDM, of whom only two had SLNB and 92% had one or more local recurrences prior to developing distant metastases.[13] By contrast, 49 patients (40%) with MDM developed a distant metastasis, 80% as the first event. Patients who had a >2 mm Breslow thickness melanoma excised with a 2-cm margin had distant spread in 3% of PDM cases with PDM and 40% of those with MDM ($P < 0.001$). Aggressive early surgical management may reduce the risk of distant metastases, although the presence of recurrence may signal a high-risk group.

This review has some limitations. The review included two instances of studies identified in the search strategy that had used the same database or population cohort. This introduced the risk of multiple publication bias, albeit by different authors. Therefore, in duplicate patient cohorts, the smaller series was excluded where there was overlap in time periods. Therefore, some patients who were not in the overlapping period were not included, thus limiting the patient pool. Where two studies were published from the same institution yet one analysed a single anatomical site only, both were included. The review should be repeated in two years as an update.

In conclusion, the rates of positive SLNB in DM (6.5%) are less compared to those in other variants of melanoma; however, nodal status may still predict DFS and MSS.[1,9,15,19,20] MDM is associated with a higher rate of micro-metastases to regional lymph nodes (13.8%) than PDM (5.4%), and DFS and MSS may be lesser in MDM than in PDM. Considering the rate of positive SLNB in the histological subtypes of DM, procedural complications and cost, we recommend considering SLNB in MDM. However, with such low rates of positive SLNB in PDM and in the absence of high-risk features to stratify patients, we would not recommend SLNB in PDM.

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Nil.

## Conflict of interest

Nil.

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